

TRACEBIND Phase 4 Protocol Specification

Empirical Validation Protocol for Observational Atmospheric Fields (ERA5 & GOES)

Protocol Version:	1.0.0 (Preregistered Methodological Validation Plan)	Target Date:	July 2026
Repository Branch:	v1.1-dev-empirical (Derived from v1.0.0-frozen)	Primary Dataset:	ERA5 Reanalysis (500 hPa Geopotential / MSLP)
Primary Question:	Does TRACEBIND capture complementary spatial phase structure not summarized by established statistics?		

1. Scope & Paper Separation

To maintain scientific discipline and prevent scope expansion, TRACEBIND validation is partitioned into two distinct paper scopes:

- Paper 1 (Methodological Validation - Current Scope):** Evaluates whether TRACEBIND measures complementary spatial phase structure beyond existing spatial metrics, texture descriptors, and phase baselines under controlled observational conditions.
- Paper 2 (Domain Science Application - Future Work):** Evaluates whether TRACEBIND improves operational meteorological forecasting, cyclone intensity prediction, or climate model diagnostics.

2. Response Variable Definition & Pre-registered Response

Explicit Response Variable Definition

The primary response variable $Y(t)$ is formally fixed prior to data evaluation as the **Cyclone Central Pressure Deficit** (ΔP in hPa) relative to standard atmospheric baseline:

$$\Delta P(t) = P_{env}(t) - P_{min}(t) = 1013.25 \text{ hPa} - P_{min}(t)$$

Secondary sensitivity check: Maximum Sustained Wind Speed (V_{max} in m/s). No additional response metrics will be evaluated to eliminate post-hoc outcome selection.

3. Expanded Baseline Metric Matrix

TRACEBIND (T) is benchmarked against eight baseline descriptors across classical spatial, domain texture, and phase-sensitive categories:

Category	Metric Name	Mathematical Designation	Primary Feature Captured
Classical Spatial	Moran's I	I	Global second-order spatial covariance
Classical Spatial	Geary's C	C	Local squared difference spatial variability
Classical Spatial	Semivariogram Range	a_0	Spatial correlation length scale
Atmospheric Domain	Spatial Variance	σ^2	Global field intensity dispersion

Category	Metric Name	Mathematical Designation	Primary Feature Captured
Atmospheric Domain	Texture Entropy	H	First-order histogram information entropy
Atmospheric Domain	Gradient Energy	$E_g = \sum \ \nabla Z\ ^2$	Local spatial gradient sharpness
Atmospheric Domain	Laplacian Energy	$E_L = \sum \nabla^2 Z $	Structural boundary and curvature density
Phase Benchmark	Phase Congruency	PC	Local frequency phase alignment
Proposed Metric	TRACEBIND Index	T	Integrated Gradient Concentration & Spectral Phase Alignment

4. Preregistered Decision Rules & Success Criteria

Preregistered Success Thresholds

TRACEBIND will be deemed to provide statistically significant complementary information if and only if all four conditions are met:

- Akaike Information Criterion Reduction:** $\Delta AIC = AIC_{Model2} - AIC_{Model1} < -2.0$
- Likelihood Ratio Test Significance:** Likelihood-ratio test $p < 0.05$ using HC3 robust covariance matrix.
- Out-of-Sample CV Predictive Gain:** Cross-validated explanatory gain $\Delta R^2_{CV} > 0.02$ under GroupKFold (grouped by independent cyclone events).
- Independent Replicability:** Effect verified across at least two independent atmospheric events (ERA5 mid-latitude cyclogenesis + GOES convective outbreak).

5. Observational Benchmarks: B7 & B8 Specifications

5.1 Benchmark 7: Predictive Lead Time & Temporal Stability

Evaluates whether TRACEBIND exhibits predictive lead time (τ) prior to rapid physical intensification transitions (t_{RI}).

- Predictive Lead Time Equation:** $\tau = t_{RI} - t_{threshold}(T)$, where threshold is set at $\mu + 2\sigma$ baseline.
- Temporal Smoothness Metric:** Evaluated via first-order difference variance $Var(\Delta M / \Delta t)$ to penalize high-frequency noise.

5.2 Benchmark 8: Quantitative Spatial Localization

Replaces subjective heatmap inspection with three objective localization statistics computed over sliding 32×32 sub-windows:

- Localization Error (d_{loc}):** Distance (km) between metric peak location and cyclone minimum pressure center.
- Vorticity Overlap (IoU):** Intersection-over-Union between top 10% metric intensity mask and top 10% relative vorticity field ζ .
- Spatial Concentration Ratio ($C_{spatial}$):** Ratio of top 10% spatial tile sum to total domain field sum.

6. Pre-defined Outcomes & Methodological Fallbacks

Outcome Scenario	Empirical Criteria	Scientific Interpretation & Action Plan
Outcome A (Supported)	$\Delta AIC < -2, p < 0.05, \Delta R^2 > 0.02$	TRACEBIND captures complementary spatial phase structure. Proceed to Paper 1 journal submission.
Outcome B (Redundant)	$\Delta AIC > -2$, aligned with Phase Congruency	TRACEBIND duplicates existing phase metrics. Publish as methodological evaluation bounding TRACEBIND scope.
Outcome C (Unstable)	High temporal noise, $Var(\Delta T/\Delta t) >> Var(\Delta I/\Delta t)$	TRACEBIND is overly sensitive to observational noise in satellite/reanalysis fields. Bounded utility established.

7. Software Versioning & Repository Lineage

v1.0.0-frozen (Synthetic Suite Frozen) → v1.1-dev-empirical (Phase 4 Protocol) → v2.0-research (Future Work)